M101 · Unit 13 · Quick reference

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Connecting mathematical representations
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Recall **Displacement**	$\Delta r = r(b) - r(a)$. Fixed axes; position in length units.
Recall **Velocity and speed**	$v = r' = (x', y')$; $s = \mid v \mid = \sqrt{v_x^2 + v_y^2}$. Perpendicular physical axes.
Understand **Tangent prediction**	$r(t + h) \approx r(t) + hv(t)$. Assess error in position units; stay within domain.
Recall **Accumulation**	$\int_a^b v\,dt = r(b) - r(a)$; add initial position for final position.
Understand **Distance**	$D = \int_a^b \mid v \mid dt \geq \mid \Delta r \mid$ for $a \leq b$. Magnitude inside integral.
Understand **Fixed calibration**	$z = Ar$, $r = Bz$, $BA = I$. Constant matrices commute with differentiation/integration.

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Consult **Baseline model**	$x = ut,\ y = wt - kt^2$; $u = 2$ m/s, $w = 4$ m/s, $k = 1$ m/s², $0 \le t \le 4$ s. Tangent error $(0,\ -kh^2)$.
Consult **Baseline recorder**	$p = x + y,\ q = x - y$; $x = (p + q) / 2,\ y = (p - q) / 2$. $\lvert z \rvert = \sqrt{2}\ \lvert r \rvert$.
Recall **Trapezoidal rule**	$h = (b - a) / n$; $Q_n = h[\frac{1}{2}f(a) + \sum_{j=1}^{n-1} f(a + jh) + \frac{1}{2}f(b)]$. Positive integer n; exact for linear f.
Understand **Transfer network**	$V_A^{'} = -q,\ V_B^{'} = q - r$; $(V_A + V_B)^{'} = -r$. Initial values, non-negative volumes and capacity matter.
Understand **Independent checks**	Endpoint differences; constant/shifted-linear integrals; fixed-calibration basis vectors; zero interval; constant velocity; symmetry.
Consult **Report**	Model/domain/units → quantity → method → independent checks → resolution/input effects → supported conclusion. State assistance.

Consult · Python

```
from math import hypot, isclose
speed = hypot(vx, vy)
close = isclose(estimate, benchmark,
                rel_tol=1e-8, abs_tol=1e-10)
# Decode this unit's two channels:
x, y = (p+q)/2, (p-q)/2
# Format only for display:
print(f"{distance:.6f} m")
```

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